

EX.No. : 6

SELECT with various clause – BETWEEN, IN, Aggregate function

AIM:

To view the records from the Saveetha Hospital table using SELECT commands with BETWEEN, IN and Aggregate Functions.

DESCRIPTION:

The BETWEEN operator tests whether a value lies within a specified range. The IN operator checks whether a value matches any value in a specified list. Aggregate functions such as AVG, COUNT, SUM, MIN and MAX perform calculations on a set of rows.

BETWEEN Operator:

SELECT column1, column2, ... FROM table_name WHERE expression [NOT] BETWEEN begin_value AND end_value;

IN Operator:

SELECT column1, column2, ... FROM table_name WHERE column_name IN ('value1','value2',...);

Aggregate Functions:

AVG() – average value

MAX() – maximum value

MIN() – minimum value

SUM() – total value

COUNT() – number of rows

Saveetha Hospital Dataset:

patients(patient_name, ward, treatment, doctor, operation, operation_fee)

Questions:

IN & BETWEEN

1. List the treatment types 'Heart Attack' and 'Accident'.
2. List the patients whose operation fees are in the range 5000 to 50000.

AGGREGATE

1. Find the average operation fee of patients who underwent an operation.
2. List the maximum and minimum operation fee for patients who underwent an operation.
3. List the maximum, minimum and average operation fee for each operation status.
4. List the treatment and average operation fee of each treatment.

5. Calculate the sum of all the operation fees.
6. How many patients are recorded for each treatment? Display the treatment and number of patients.
7. How many treatments did each patient register for?

OUTPUTS: IN & BETWEEN

1. List the treatment types 'Heart Attack' and 'Accident'.

```
mysql> select treatment from patients where treatment in ('Heart Attack','Accident');

+-----+
| treatment |
+-----+
| Heart Attack |
| Accident |
+-----+
2 row(s) in set (0.00 sec)

mysql>
```

2. List the patients whose operation fees are in the range 5000 to 50000.

```
mysql> select patient_name,operation_fee from patients where operation_fee between 5000 and 50000;

+-----+-----+
| patient_name | operation_fee |
+-----+-----+
| Aishu | 5000 |
| Arun | 10000 |
| Raja | 10000 |
| Ashitha | 50000 |
+-----+-----+
4 row(s) in set (0.00 sec)

mysql>
```

OUTPUTS: AGGREGATE

AGGREGATE

1. Find the average operation fee of patients who underwent an operation.

```
mysql> select avg(operation_fee) from patients where operation='Yes';

+-----+
| avg(operation_fee) |
+-----+
| 72000.0 |
+-----+
1 row(s) in set (0.00 sec)

mysql>
```

2. List the maximum and minimum operation fee for patients who underwent an operation.

```
mysql> select max(operation_fee),min(operation_fee) from patients where operation='Yes';
```

```
+-----+-----+
| max(operation_fee) | min(operation_fee) |
+-----+-----+
| 100000             | 10000              |
+-----+-----+
1 row(s) in set (0.00 sec)
```

```
mysql>
```

3. List the maximum, minimum and average operation fee for each operation status.

```
mysql> select operation,max(operation_fee),min(operation_fee),avg(operation_fee) from patients group by operation;
```

```
+-----+-----+-----+-----+
| operation | max(operation_fee) | min(operation_fee) | avg(operation_fee) |
+-----+-----+-----+-----+
| Yes      | 100000             | 10000              | 72000.0            |
| No       | 10000              | 1000               | 3600.0              |
+-----+-----+-----+-----+
2 row(s) in set (0.00 sec)
```

```
mysql>
```

4. List the treatment and average operation fee of each treatment.

```
mysql> select treatment,avg(operation_fee) from patients group by treatment;
```

```
+-----+-----+
| treatment | avg(operation_fee) |
+-----+-----+
| Heart Attack | 100000.0          |
| Accident   | 100000.0          |
| BP         | 5000.0            |
| Heart      | 10000.0           |
| Tyroid     | 10000.0           |
| Cold       | 1000.0            |
| Fever      | 1000.0            |
| Lung Cancer | 50000.0           |
| Cancer     | 100000.0          |
| Stomach Pain | 1000.0            |
+-----+-----+
10 row(s) in set (0.00 sec)
```

```
mysql>
```

5. Calculate the sum of all the operation fees.

```
mysql> select sum(operation_fee) from patients;
```

```
+-----+
| sum(operation_fee) |
+-----+
| 378000             |
+-----+
1 row(s) in set (0.00 sec)
```

```
mysql>
```

6. How many patients are recorded for each treatment? Display the treatment and number of patients.

```
mysql> select treatment,count(patient_name) from patients group by treatment;
```

```
+-----+-----+
| treatment | count(patient_name) |
+-----+-----+
| Heart Attack | 1 |
| Accident | 1 |
| BP | 1 |
| Heart | 1 |
| Tyroid | 1 |
| Cold | 1 |
| Fever | 1 |
| Lung Cancer | 1 |
| Cancer | 1 |
| Stomach Pain | 1 |
+-----+-----+
10 row(s) in set (0.00 sec)
```

```
mysql>
```

7. How many treatments did each patient register for?

```
mysql> select patient_name,count(treatment) from patients group by patient_name;
```

```
+-----+-----+
| patient_name | count(treatment) |
+-----+-----+
| Ajay | 1 |
| Tharun | 1 |
| Aishu | 1 |
| Arun | 1 |
| Raja | 1 |
| Harish | 1 |
| Vadivu | 1 |
| Ashitha | 1 |
| Kamini | 1 |
| Balaji | 1 |
+-----+-----+
10 row(s) in set (0.00 sec)
```

```
mysql>
```

RESULT:

The records from the Saveetha Hospital table are successfully retrieved using SELECT commands with BETWEEN, IN and aggregate functions.